Rational Number Project

Initial Fraction Ideas

Abridged edition for use with third graders

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The original authors of the RNP fractions lessons are, Kathleen Cramer, Merlyn Behr, Thomas Post, and Richard Lesh. Contributors to this revised module for use with third graders include, Dr. Terry Wyberg of the University of Minnesota, Dr. Susan Ahrendt, University of Wisconsin, River Falls, Dr. Debra Monson, St. Thomas University and Christina Miller, Phd student at the University of Minnesota. We are also grateful to the third grade teachers from Minneapolis Public Schools for their ideas for revising the original RNP lessons to better meet the needs of third graders.

ABOUT THE AUTHORS for the Original RNP Fraction Lessons

Kathleen Cramer is an Associate Professor in the College of Education and Human Development at the University of Minnesota. She teachers graduate and undergraduate mathematics education classes for students majoring in elementary education. Her research interests continue to focus on better understanding how to help elementary aged students build meaning for rational numbers and proportionality. Kathleen has a Ph.D. from the University of Minnesota in mathematics education. She has published articles and book chapters dealing with the teaching and learning of fractions and proportional reasoning. She has done numerous workshops for teachers dealing with fraction instruction.

Professor Cramer has been involved with the Rational Number Project (RNP) since 1980. She participated in the initial teaching experiments with fourth and fifth graders. She has taken the primary responsibility for revising the lessons developed from the research to form the two sets of RNP Fraction Curriculum Modules.

Merlyn Behr was for over 25 years a professor of mathematics education at Northern Illinois University in DeKalb, Illinois. He was also a faculty member at Florida State University where he received his Ph.D., and at Louisiana State University at Baton Rouge. Merlyn's primary interest was in children's learning of elementary- and middle-grades mathematical concepts. He contributed a great deal to our understanding of children's cognitive processes in these areas. He was very active in the research community and served on the editorial board of the Journal for Research in Mathematics Education (JRME) and as chair of the North American chapter of the research group of the Psychology of Mathematics Education. As a co-founder of the RNP, Merlyn was instrumental in charting its course and providing much valued intellectual leadership in many aspects of RNP activity.

Merlyn died in February 1995. His wit and professional contributions are sorely missed.

Thomas Post, former high school mathematics teacher in New York State, joined the faculty of the College of Education at the University of Minnesota in 1967 after receiving his Ed.D. from Indiana University. Professor Post's interest is closely allied with other RNP members, as he is especially interested in children's and teachers' perceptions of middle-school mathematics. He also has an interest in interdisciplinary approaches to curriculum. He was a co-founder of the RNP and has been active in the mathematics education research community. Along with Kathleen Cramer, Merlyn Behr and Richard Lesh, he has been one of the co-authors of some 70 papers, book chapters and technical reports produced by the RNP since it's inception in 1979. Tom has also served on the editorial board of the JRME and has been chair of the North American chapter of the research group Psychology of Mathematics Education.

Richard Lesh, former professor and dean at Northwestern University, received his Ph.D. from Indiana University. He spent 5 years overseeing computer software development in mathematics and science at WICAT systems in Provo, Utah. He then served as senior research scientist at ETS in Princeton, NJ where he developed innovative strategies and materials for assessing outcomes in mathematics classrooms. Professor Lesh has served as project manager of the program unit - Research on Teaching and Learning - at the National Science foundation. Currently, he is a professor of mathematics at the University of Massachusetts-Dartmouth helping to further advance our thinking about authentic assessment, principles and strategies. Dick is on of the original co-founders of the RNP and has worked on each of its six grants since 1979. He currently leads the Massachusetts site of the RNP's Middle-Grades Teacher Enhancement Project, which is the latest of the projects funded by NSF.

Content

Preface

I. Teacher's Guide

- Background for the original RNP lessons
- The RNP Curriculum
- Theoretical Framework
- Lesson Format
- Manipulative Materials
- Number Line for fractions
- Special Notes on Children's Thinking
- Final Comments

II. The RNP Lessons: Initial Fraction Ideas: Abridged version for using with third graders

- Detailed Scope and Sequence

- Lesson 1: Exploring with Fraction Circles
- Lesson 2: Modeling with Fraction Circles
- Lesson 2.5: Modeling with Fraction Circles
- Lesson 3: Modeling with Fraction Circles
- Lesson 4: Comparing Paper Folding and Circles
- Lesson 5: Connecting Models with Symbols
- Lesson 6: Comparing Unit Fractions
- Lesson 6.5: Comparing fractions with like numerators and like denominators
- Lesson 7: Paper Folding and Comparing Fractions
- Lesson 8: Fraction Circles and Equivalence
- Lesson 9: Fraction Circles, Paper Folding and Equivalence
- Lesson 10: Paper Folding and Equivalence
- Lesson 11: Comparing Fractions to One-half
- Lesson 12: Introducing the Chip Model
- Lesson 13: Using the Chip Model
- Lesson 14: Fraction Circles and Fractions Greater than One
- Lesson 15: Number line and whole numbers
- Lesson 16: Context, paper folding and number lines
- Lesson 17: Translating from paper folding to a number line
- Lesson 18: Translating from fraction circles to a number line

Preface

The **Rational Number Project** (RNP) is a cooperative research and development project funded by the National Science Foundation. Project personnel have been investigating children's learning of fractions, ratios, decimals and proportionality since 1979. The original book of fraction lessons was the product of several years of working with children in classrooms as we tried to understand how to organize instruction so students develop a deep, conceptual understanding of fractions.

The lessons were originally published in 1997 under the title: RNP: *Fraction Lessons for the Middle Grades Level 1*. They have been revised (August, 2009) and renamed to better reflect its content. This module called, *Initial Fraction Ideas,* includes lessons to develop meaning for fractions using a part-whole model, to support students' construction of informal ordering strategies based on mental representations for fractions, to build meaning for equivalence concretely. The lessons also develop meaning for adding and subtracting fractions using concrete models. *Initial Fraction Ideas* module does not include formal algorithms, and instruction with formal algorithms was not part of this original RNP curriculum module.

This edition includes a subset of the 23 lessons found in the *RNP: Initial Fraction Ideas* curriculum. The lessons selected and subsequently revised for third graders reflects the third grade Common Core Standards for fractions as well as the Minnesota State Standards for fractions at this grade level. The revisions are based on input from third grade teachers who used the original RNP lessons in their classrooms. Four additional lessons dealing with the number line as a

model for fractions were created and field tested with third graders and revised based on that experience. The number line is a challenging model for young learners but lessons were created as the Common Core Standards and MN State Standards both include the number line model in their third grade standards. The third grade module consists of 20 lessons in all.

The complete set of 23 fraction lessons from the module, *Initial Fraction Ideas,* can be found at:
http://www.cehd.umn.edu/ci/rationalnumberproject/rnp1-09.html

A companion module to this module has been developed with NSF support. This module, *Fraction Operations and Initial Decimal Ideas*, extends students' fraction ideas to develop fraction operations of addition, subtraction, multiplication and division with symbols. That module also introduces students to decimal ideas – naming decimals, order, equivalence, addition and subtraction. This module can be found on the RNP website at this address:

http://www.cehd.umn.edu/ci/rationalnumberproject/rnp2.html

- **This revised set of lessons** provide teachers with an alternative to the textbook scope and sequence for fraction instruction and are appropriate for students in grade 3 but will be effective in remedial settings with older students.

- **This revised set of lessons** help students develop number sense for fractions because they invest time in the development of concepts, order and equivalence ideas.

- **This revised set of lessons** provides students with daily "hands-on" experiences. Fraction circles, chips and paper folding are the manipulative models used in these lessons to develop initial fraction ideas.

- **This revised set of lessons** provides teachers with daily activities that involve children in large group and small group settings. All the lessons involve students using manipulative materials. Our work with children has shown that students need extended periods of time with manipulatives to develop meaning for these numbers.

- **This revised set of lessons** offers teachers insight into student thinking as captured from the RNP research with children. The "Notes to the Teacher" section shares examples of students' misunderstandings, provides anecdotes of student thinking, and contains information on using manipulative materials.

- **These lessons** will help teachers and students attain the goals set forth in the Common Core Standards for grade 3.

Grade Level	**CCSS Objective**	**RNP 1 Lesson #**
3	**Develop understanding of Fractions as numbers**	
	1. Understand a fraction 1/b as a quantity formed by 1 part when a whole is partitioned into b equal parts; understand a fraction a/b as the quantity formed by a parts of size b	1-5; 12, 13
	2. Understand a fraction as a number on a number line; represent fractions on a number line diagram.	15-18

	a. Represent a fraction 1/b on a number line diagram by defining the interval between 0 and 1 as the whole and partitioning it into b equal parts Recognize that each part has the size 1/b and that the endpoint of the part based at 0 locates the number 1/b on the number line.	15-18
	b. Represent a fraction a/b on a number line diagram by marking off a length of 1/b from 0. Recognize that the resulting interval has size a/b and that its endpoint locates the number a/b on the number line.	15-18
	3. Explain equivalence of fractions in special cases, and compare fractions by reasoning about their size.	8 – 10
	a. Understand two fractions are equivalent (equal) if they are the same size or the same point on the number line	8 – 10 (Not with # line model)
	b. Recognize and generate simple equivalent fractions. Explain why the fraction are equivalents by using a visual model	8 – 10
	c. Express whole numbers as fractions, and recognize fractions that are equivalent to whole numbers.	14
	d. Compare two fractions with the same numerator or the same denominator by reasoning about their size. Recognize comparisons are valid only when the two fractions refer to the same whole. Record results of comparisons with symbols >, <, = and justify conclusions by using a visual fraction model.	6, 6.5, 7, 11

- **These lessons** will help teachers and students attain the goals set fort in the Minnesota Mathematics Standards for grade 3 and expose students to fraction equivalence ideas found in the grade 4 standards.

Standard	**Benchmark (3rd Grade)**	**RNP lesson(s)**
Understand meanings and uses of fractions in real-world and mathematical situations	Read and write fractions with words and symbols. Recognize that fractions can be used to represent parts of a whole, parts of a set, points on a number line, or distances on a number line.	1-5; 12-14; 15-18
	Understand that the size of a fractional part is relative to the size of the whole.	1 - 5
	Order and compare unit fractions and fractions with like denominators by using models and an understanding of the concept of numerator and denominator.*	6, 6.5, 7, 11
Standard	**Benchmark (4th grade)**	
Represent and compare fractions and decimals in real-world and mathematical situations; use place value to understand how decimals represent quantities	Represent equivalent fractions using fraction models such as parts of a set, fraction circles, fraction strips, number lines and other manipulatives. Use models to determine equivalent fractions.	8-11

- Order lessons also include comparing non-unit fractions with same numerator and exploration of ½ as a benchmark.

Teacher's Guide
RNP: Initial Fraction Ideas

Abridged edition for use with third graders

Background of the Original RNP lessons

The original RNP lessons reflect research on children's fraction learning conducted by the National Science Foundation-sponsored Rational Number Project (RNP). Since 1978, Merlyn Behr (Northern Illinois University), Kathleen Cramer (University of Minnesota), Thomas Post (University of Minnesota) and Richard Lesh (Indiana University) have studied how elementary-aged children learn to extend their understanding of numbers to include fraction ideas.

The RNP staff conducted teaching experiments with fourth- and fifth-grade children. In a teaching experiment, researchers entered the classroom as teachers and worked with a curriculum created from a well thought-out theoretical framework. During the teaching and learning process, researchers study children's learning as they progress through the lessons. This is done through classroom observations, student interviews and written assessments.

The curriculum created for the teaching experiments that evolved into these lessons reflected the following beliefs: (a) Children learn best through active involvement with multiple concrete models, (b) physical aids are just one component in the acquisition of concepts: verbal, pictorial, symbolic and real-world representations also are important, (c) children should have opportunities to talk together and with their teacher about mathematical ideas, and (d) curriculum must focus on the development of conceptual knowledge prior to formal work with symbols and algorithms.

The teaching experiments were conducted in two parts. The first phase of the project was ten weeks long with small groups of fourth and fifth graders at two different sites (Minnesota and Illinois). The second phase was conducted with a classroom of 30 students. A class of fourth graders participated from January of their fourth grade, through January of their fifth grade. Instruction was four days per week and covered the following topics: part-whole model for fractions, ratio and quotient models for fractions, decimals, and number lines.

Throughout both teaching experiments a subset of children were interviewed every two weeks. The interviews provided information on children's thinking about fraction ideas. We were interested in what role manipulative materials played in their thinking as well as what understandings and misunderstandings children have with fractions.

Our work with children helped explain why children have so much difficulty with fractions. It also informed us as to the type of experiences children need to develop a deep, conceptual understanding of fractions. Consider a few of the insights garnered from these teaching experiments:

1. Children have difficulty internalizing that the symbol for a fraction represents a single entity. When asked if $\frac{2}{3}$ was one or two numbers, many children would say that the symbol represented two numbers. When students consider $\frac{2}{3}$ as two numbers then it makes sense to treat them like whole numbers. For example, when students add two fractions by adding the numerators and then denominators, they are interpreting the symbols as four numbers, not two. Many errors with fractions can be traced to students' lack of mental images for the quantity the symbol represents.

2. Ordering fractions is more complex than ordering whole numbers. Comparing $\frac{1}{4}$ and $\frac{1}{6}$ conflicts with children's whole number ideas. Six is greater than four, but $\frac{1}{4}$ is greater than $\frac{1}{6}$. With fractions, **more** can mean **less**. The **more** equal parts you partition a unit into, the **smaller** each part becomes. In contrast, $\frac{3}{5}$ is greater than $\frac{2}{5}$ because 3 of the same-size parts are greater than 2 of the same-size parts. In this case, **more** implies **more**. Being able to order plays an important part in estimating fraction addition and subtraction. Ideally when a student adds, for example, $\frac{1}{4}+\frac{1}{3}$, she should be able to reason from her mental images of the symbols that (a) the answer is greater than $\frac{1}{2}$, but less than one and (b) $\frac{2}{7}$ is an unreasonable answer because it is less than $\frac{1}{2}$.

3. Understanding fraction equivalence is not as simple as it may seem. Some children have difficulty noting equivalence from pictures. Imagine a circle partitioned into fourths with one of those fourths partitioned into three equal parts. Some children we worked with were unable to agree that $\frac{3}{12}$ equals $\frac{1}{4}$ even thought they agreed that physically the two sections were the same size. Children said that once the lines were drawn in, you could not remove them. [Therefore $\frac{3}{12} \neq \frac{1}{4}$]. In reality, that is just what must be done to understand fraction equivalence from a picture.

4. Difficulties children have with fraction addition and subtraction come from asking them to operate on fractions before they have a strong conceptual understanding for these new numbers. They have difficulty understanding why common denominators are needed so they revert to whole number thinking and add numerators and denominators.

The RNP Curriculum

The RNP curriculum offers an alternative scope and sequence to one suggested in fourth- or fifth-grade textbooks. The RNP philosophy is that extended periods of time invested with manipulative materials developing concepts, order, and equivalence ideas are needed before students can operate on fractions in a meaningful way. We call these skills, initial fraction ideas. These goals are consistent with the instructional goals set forth in the National Council of Teachers of Mathematics in their *Principles and Standards for School Mathematics*. The RNP curriculum provides teachers with carefully researched lessons to meet these goals.

The RNP Level 1 materials develop the following topics: (a) part-whole model for fractions, (b) concept of unit, (c) concepts of order and equivalence and (d) addition and subtraction of fractions at the concrete level. The concrete models used are fraction circles, paper folding and chips. It de-emphasizes written procedures for ordering fractions, finding fraction equivalences, and

symbolic procedures for operating on fractions. Instead it emphasizes the development of a quantitative sense of fraction.

To think quantitatively about fractions, students should know something about the relative size of fractions and be able to estimate a reasonable answer when fractions are operated on. Below, find an example of a fourth-grad student's thought process for estimating a fraction addition problem. This student used the RNP curriculum; her thinking reflects a quantitative sense of fraction. Students using the RNP lessons develop this type of understanding for fractions.

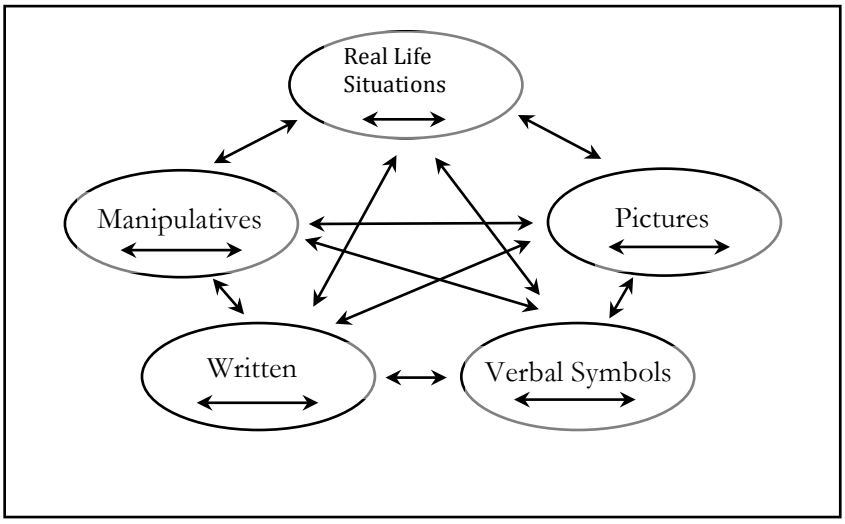
> **Problem: John calculated the problem as follows: 2/3 + 1/4 = 3/7. Do you agree?**
>
> *Student*: I don't agree. He did it weird. You don't add the top numbers and bottom numbers.
>
> *Teacher*: What would be an estimate?
>
> *Student*: It would be…greater than 1/2 because 2/3 is greater than 1/2.
>
> *Teacher*: Would it be greater or less than one?
>
> *Student*: Less than one. You'd need 1/3 and 1/4 is less than 1/3.
>
> *Teacher*: What about 3/7?
>
> *Student*: 3/7 is less than 1/2.
>
> *Teacher*: How do you know?
>
> *Student*: Because 3/7 isn't 1/2. I just know.

State and National mathematics Standards have moved fraction content that in the past was fourth grade content to third grade. This abridged version of the RNP level 1 materials was developed based on third grade teachers' feedback who used a subset of the original RNP level 1 materials in their classrooms and covers content for naming fractions, ordering fractions and fraction equivalence. In addition 3 new lessons were developed to introduce students to the number line model for fractions.

Theoretical Framework

Children using these lessons will be using several manipulative models and will consider how these models are alike and different. They will work in small groups talking about fraction ideas as well as interacting with the teacher in large group settings. They will be drawing pictures to record their actions with fraction models. They will be solving story problems using manipulatives to model actions in the stories.

This model for teaching and learning reflects the theoretical framework suggested by Jean Piaget, Jerome Bruner, and Zoltan Dienes. Richard Lesh, a long time RNP member, suggested an instructional model that clearly shows how to organize instruction so children are actively involved in their learning. Consider this picture.



Lesh suggests that mathematical ideas can be represented in the five ways shown here. Children learn by having opportunities to explore ideas in these different ways and by making connections between different representations. This model guided the development of the RNP curriculum.

Lesson Format

The lessons reflect a classroom organization that values the important role a teacher plays in student learning as well as the need for students to work cooperatively, talking about ideas, and using manipulative models to represent

rational number concepts. Each lesson includes an overview of the mathematical idea developed. Materials needed by teachers and students are noted. The lesson begins with a class **Warm Up**. Warm Ups are used to review ideas developed in previous lessons and should take only 5-10 minutes of class time. There is a **Large Group Introduction** section in each lesson. The teacher's lesson plans provide problems and questions to generate discussion and target the exploration. **Small Group/Partner Work** is included in each lesson where students together continue the exploration of ideas introduced in the large group. The class ends with a **Wrap Up**. A final activity is presented to bring closure to the lesson. At times this will be a presentation by students of select problems from the group work. We found that students like to share their thinking. At other times the Wrap Up will be another problem to solve as a group. The amount of time needed for each lesson will vary from classroom to classroom. A single lesson does not necessarily reflect one day's work, though teachers often will find that one day is sufficient to cover the material.

An important part of each lesson is the "Comments" section. Here insights into student thinking captured from the initial RNP teaching experiments are communicated to teachers. These notes clarify a wide variety of issues, such as why mastery at the symbolic level is not the primary objective for many of the earlier lessons. The notes also share examples of students' misunderstandings for teacher's reflection and anecdotes of student thinking from earlier RNP projects. These notes to the teachers also clarify methods for using manipulative materials to model fraction ideas.

Manipulative Materials

Fraction circles, two-sided colored counters and paper folding are the manipulative models used. Our research has shown that the fraction circles are the most important manipulative models for developing mental images of fraction symbols.

Fraction Circles

The master for the fraction circles are in the appendix with a page showing the different partitions and colors used for the fraction circles. The circles should be duplicated on index using colors noted on each master. Teachers who have used the fraction circles have relied on their students, parents or teacher-aids to cut out the circles and to organize them in two-pocket folders. If you choose to send home the fraction circles to be cut out with the parent's help, you will find in the appendix a parent and child activity sheet for them to do together once the circles are cut out.

Counters

Two-sided colored counters are available from most publishers of mathematics manipulative materials. A less expensive way is to purchase from a tile store, one square inch tiles (white on one side, tan on other). These cost less than 1.5 cents per tile. Thirty per student should be enough.

Paper Folding

Use 8.5″ by 11″ sheets of paper cut into strips 1″ by 8.5″. Have lots on hand for students to use for lessons 7 and 10.

Number line for fractions

The number line is considered to be an important model for fractions even though it is a more abstract one than other concrete and pictorial models used in fraction instruction. But this model is unique as compared to other models used to teach fractions and is more challenging for students. The unit on the number line is a length and totally continuous with no separation between units. This makes if difficult for students to identify the unit on the number line. The number line uses symbols and visual cues to convey part of its meaning to

students. Students have to coordinate symbolic and visual cues to bring meaning to the model. This coordination is not needed with other more concrete models. Students also struggle interpreting the tick marks (partitions) on the number line often counting the tick marks and not the distances to identify fractions. The number line lessons in the third grade edition of the RNP lessons tries to address the challenges students have with this model for fractions. Context and translations from paper folding to the number line are two strategies used to build meaning for the number line as a model for fractions. We are continuing to study how to effectively build this model into the RNP lessons. The 3 lessons currently available are working drafts. These will be updated periodically as we implement them ourselves with students to study student's thinking related to the number line and evaluate the effectiveness of the lessons with 3rd graders.

Special Notes on Students' Thinking

From our interviews with children we noted that they constructed what we now refer to as informal strategies for ordering fractions. These strategies reflect students' use of mental images of fractions to judge the fraction's relative size. These informal strategies do not rely on procedures usually taught: least common denominators and cross-products. We have named the four strategies noted in students' thinking as: same numerator, same denominator, transitive and residual strategies.

When comparing $\frac{2}{3}$ and $\frac{2}{6}$ (fractions with the same numerator) students can conclude that $\frac{2}{3}$ is the larger fraction because thirds are larger than sixths and two of the larger pieces must be more than two of the smaller pieces. This

strategy involves understanding that an inverse relationship exists between the number of parts a unit is partitioned into and the size of the parts.

The same denominator strategy refers to fractions like $\frac{3}{8}$ and $\frac{2}{8}$. In this case, the same denominator implies that one is comparing parts of the unit that are the same size. Three of the same-size parts are greater than two of the same-size parts.

The student strategy that has been termed the *transitive* strategy can be modeled by comparing $\frac{3}{7}$and $\frac{5}{9}$. When making this comparison, a student can conclude that $\frac{3}{7}$ is less than $\frac{5}{9}$ because $\frac{3}{7}$ is less than $\frac{1}{2}$, while $\frac{5}{9}$ is greater than $\frac{1}{2}$. This is the transitive strategy because students use a single outside value to compare both fractions.

When comparing $\frac{3}{4}$ and $\frac{5}{6}$, a student can reflect that both fractions are one "piece" away from the whole unit. Because $\frac{1}{6}$ is less than $\frac{1}{4}$, $\frac{5}{6}$ must be closer to the whole and is therefore the bigger fraction. This thinking strategy has been called a *residual* strategy because students focus on the part "leftover" in judging the relative size of the fractions.

These four strategies closely parallel students' actions with manipulatives. They are in contrast to the paper and pencil procedures, which require changing both fractions to common denominators or calculating cross-products. RNP lessons developed only these student-constructed strategies. The order questions on the interviews will assess whether students construct these strategies. Students who have constructed these strategies have developed are on the way to developing number sense for fractions.

Final Comments

You will find at the end of each lesson a form for you to record your adaptations for each lesson. Any curriculum will need to be "personalized" by the teacher who uses it, so it best meets the needs of his/her students. This form

will act as a reminder about changes you feel are important to make the next time you teach the lesson.

The RNP Lessons
Initial Fraction Ideas

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Scope and Sequence

RATIONAL NUMBER PROJECT
Initial Fraction Ideas
Scope and Sequence

LESSON	MANIPULATIVE	TOPIC
1	Fraction Circles	Exploration with the circles.
2	Fraction Circles	Model and verbally name: 1-half, 1-third
2.5	Fraction Circles	Model and verbally name: 1-half, 1-third, and 1-fourth
3	Fraction Circles	Model and verbally name unit fractions with denominators greater than 4.
4	Paper Folding	Compare paper folding to fraction circles. Model and name (verbally and with written words) unit and non-unit fractions.
5	Fraction Circles	Model fractions and record with symbols a/b.
6	Fraction Circles	Model the concept that the greater the number of parts a unit is divided into, the smaller each part is.
6.5	Fraction Circles	Order fractions with like numerators and like denominators embedded in division story problems.
7	Paper Folding	Reinforce the concept that the greater the number of parts a unit is divided into, the smaller each part is.
8	Fraction Circles	Fraction Equivalence
9	Fraction Circles	Fraction Equivalence
10	Paper Folding	Fraction Equivalence.
11	Fraction Circles	Order fractions by comparing to 1-half.

12	Chips	Introduce new model for fractions less than one by comparing to a familiar model.
13	Chips	Model fractions using several units for the same fraction.
14	Fraction Circles	Model fractions greater than one using mixed and improper fraction notation.
15	Number Line	Review characteristics of a number line with whole numbers
16	Context, Paper folding and Number lines	Model fractions using pictures of paper folding strips as a model for number line. Problems embedded in story contexts connected to lengths
17	Number lines	Make connections between paper folding model and the number line
18	Number lines	Make connections between fraction circle model and the number line

Post Lesson Reflection

Lesson_____

1) Number of class periods allocated to this lesson: _____

2) Student Pages used: _____

3) Adaptations made to lesson: (For example: added extra examples, eliminated certain problems, changed fractions used)

4) Adaptations made on Student Pages:

5) The next time I teach this lesson I should: